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ORIGINAL ARTICLE

Blast waves from cylindrical charges

C. Knock · N. Davies

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Abstract Comparisons of explosives are often carried out using TNT equivalency which is based on data for spherical charges, despite the fact that many explosive charges are not spherical in shape, but cylindrical. Previous work has shown that it is possible to predict the over pressure and impulse from the curved surface of cylindrical charges using simple empirical formulae for the case when the length-to-diameter (L/D) ratio is greater or equal to 2/1. In this paper, by examining data for all length-to-diameter ratios, it is shown that it is possible to predict the peak over pressure, P, for any length-to-diameter ratio from the curved side of a bare cylindrical charge of explosive using the equation $P = K_P M(L/D)^{1/3}/R^3$, where M is the mass of explosive, R the distance from the charge and K_P is an explosive-dependent constant. Further out where the cylindrical blast wave 'heals' into a spherical one, the more complex equation $P = C_1(Z'')^{-3} + C_2(Z'')^{-2} + C_3(Z'')^{-1}$ gives a better fit to experimental data, where $Z'' = M^{1/3}(L/D)^{1/9}/D$ and C_1, C_2 and C_3 are explosive-dependent constants. The impulse is found to be independent of the L/D ratio.

Keywords Cylindrical charges · Blast · L/D ratio

1 Introduction

It has long been known [1,2] that the blast wave from an explosive depends on the explosive's shape. In 1948, Stoner and Bleakney [1] demonstrated how peak over pressure (that is the peak pressure above ambient pressure) varied between spherical, cylindrical and block charges. More recently in 1996, Victor [3] pointed out that 'for non-spherical charges, there are time and directional/orientational effects that can yield multiples of greater than 2.0′ when blast parameters are compared to those of spherical charges of the same mass. Yet although most military charges have a cylindrical based shape, safety cases and explosive comparisons are based on spherical charges and there has been little published work on the blast from cylindrical charges.

A variety of definitions are used in the literature. For clarity the following definitions will be used, with information from other authors' work adjusted accordingly. The scaled distance, Z, is defined as $Z = R/M^{1/3}$ where R is distance from the charge and M is the mass of the charge. Angles around the cylindrical charge are given as Fig. 1; 0° is the axial end from which initiation takes place, 90° is prependicular to the curved end of the charge and 180° is the axial direction at the far point from initiation.

Unlike a spherical charge where the blast wave is the same in all directions, the pattern of blast waves from a cylindrical charge is complex as shock waves radiate out from both the ends and sides of the cylinder (Fig. 1), reflecting off each other and creating bridging waves where the two shock waves meet.

Literature surveys carried out by the authors on cylindrical charges studying peak over pressure [4] and impulse [5] highlighted the limited data available for cylindrical charges. The only raw data, as opposed to fitted data in, for example, Swisdak [6], being that from Wisotski and Snyer for Composition B [7] and from Plooster for Pentolite [8].

To predict the peak over pressure from cylindrical, spherical and cuboid explosive charges Stoner and Bleakney [1] used an equation of the form $\Pi = AZ_n^{-1} + BZ_n^{-2} + CZ_n^{-3}$,

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C. Knock (✉) · N. Davies
Cranfield University, Shrivenham, Swindon, Wilts SN6 8LA, UK
e-mail: c.knock@cranfield.ac.uk

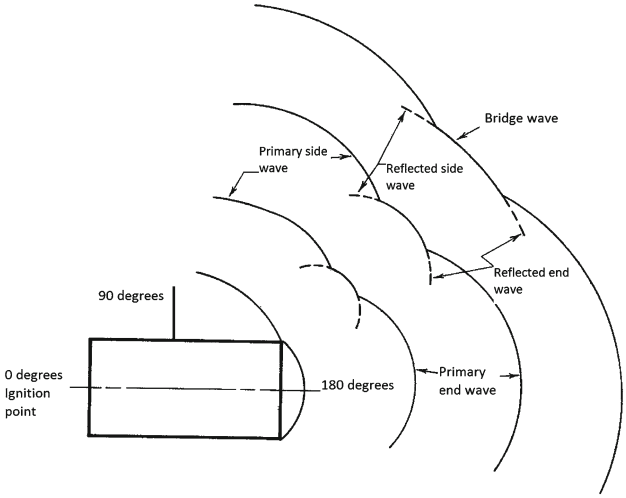


Fig. 1 Development of shock waves and bridge waves from cylindrical charges. From Wisotski and Snyer [7]. Detonation is at the end marked 180°

where A, B and C are constants obtained from least square fits of the equation to experimental data, $Z_n = R/(\rho V)^{1/3}$ where V is the volume of the charge and ρ is the specific gravity and $\Pi = (P - P_o)/P_o$ where P is the peak overpressure and P_o is the atmospheric pressure. Stoner and Bleakney explain the equation as representing a combination of the asymptotic solutions of the propagation equations, Z^{-1} and experimental measurements in general pointing to a limiting factor of Z^{-3} as a possible limiting value for the dependency at small distances. This dependency was later confirmed in the numerical solution of the fluid flow equations for strong shock (>1 MPa), point source, spherical charges [9]. The term in Z^{-2} provides a third factor 'necessary for fitting the data correctly'[1].

Other equations and graphs have been produced for predicting the peak over pressure from the curved surface of a cylinder, see literature survey and results of work in Knock and Davies [4]. However, though some authors [7,8,10] mention the dependency of blast on length-to-diameter ratio, few papers produce equations to account for this effect.

There are three surfaces to a cylinder, the curved surface and the two flat ends. Wistoksi and Snyer [7] and more recently Simeons [10] found that the peak over pressure from the curved surface of the cylinder 'decreased with a decrease in L/D ratio' [7], where L and D are the charge's length and diameter, respectively. Considering the two surfaces of a cylinder, the curved surface and the flat end, Plooster [8] found that 'the peak pressure and impulse close to the charge are largest in the direction of largest presented area', where the presented area is the surface area of the charge that can be seen at a given point. Wistoksi and Snyer found that the optimum peak over pressure from the curved surface occurred for an L/D of 6/1.

Rice and Ginell (quoted in [11]) suggested that a comparison of cylindrical charges with spherical charges can be made by using a spherical charge of the same weight giving

$$r_{\text{sph}} = (3L_{\text{cyl}}/4r_{\text{cyl}})^{1/3} r_{\text{cyl}} \tag{1}$$

where L_{cyl} and r_{cyl} are the length and radius of the charge, respectively. They found that the transition from cylindrical to a spherical case occurred at a standoff distance, that is the distance from the explosion, of approximately half the charge length.

One equation that does take into account the length-to-diameter (L/D) ratio to predict the peak over pressure for cylinders measured normally to the charge axis is [12] (cited in [3])

$$\frac{P_{\text{cylinder}}}{P_{\text{sphere}}} = \frac{5.53(L/D)^{0.308}}{Z'^{0.998}} + \frac{(Z' - L/D)}{24.99} \tag{2}$$

where Z' is given in [3] as $Z' = R/(EWF \times W)^{1/3}$ where W is the weight of TNT and EWF is the equivalent weight factor that is the ratio of the weight of TNT that will give the same curves to the weight of the explosive of interest.

Working with the Composition B data from Wisotski and Snyer, Plooster [13] developed two equations for cylindrical charges (for $L/D \geq 1$ and $L/D < 1$) to predict the peak over pressure at any angle around the charge. These equations produce a good prediction of the blast, but are explosive dependent and complex with multiple parameters for each explosive.

Considering only $L/D \geq 2/1$, Knock and Davies [4,5] developed empirical equations to predict the peak over pressure and impulse from the curved surface of a cylindrical bare charge of explosive. They used the published data for Composition B [7] and Pentolite [8] and their own data for PE4. For a scaled distance of less than 3.5 m kg$^{-1/3}$ the peak over pressure, P, is given by an equation of the form

$$P = K\frac{M}{R^3} \tag{3}$$

where K is an explosive-dependent constant; 2,695 Pa m^3 kg^{-1} for Composition B, 2,498 Pa m^3 kg^{-1} for Pentolite and 2,565 Pa m^3 kg^{-1} for PE4. This was found to give a better fit [4] to the experimental data than (1) recommended by Rice and Ginell.

Further out, at scaled distances, $Z > 3.5$ m kg$^{-1/3}$, when the blast field begins to 'heal' to that of a sphere Knock and Davies [4] found that an equation of the form used by Stoner and Bleakney [1]

$$P = K_1\frac{1}{Z^3} + K_2\frac{1}{Z^2} + K_3\frac{1}{Z} \tag{4}$$

gives a good fit for all distances, where K_1, K_2 and K_3 are explosive-dependent constants.

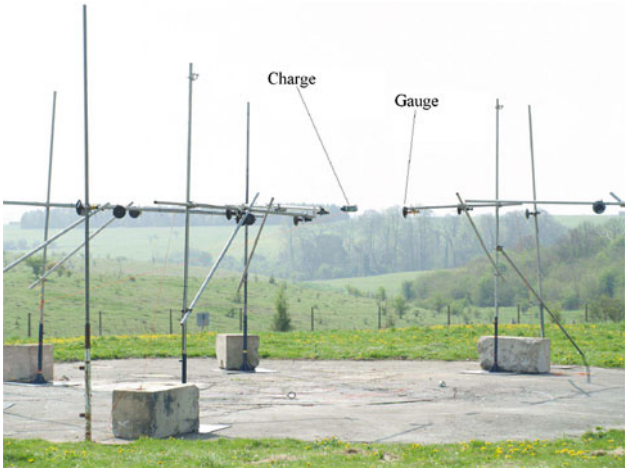


Fig. 2 Experimental setup for cylindrical charges of PE4

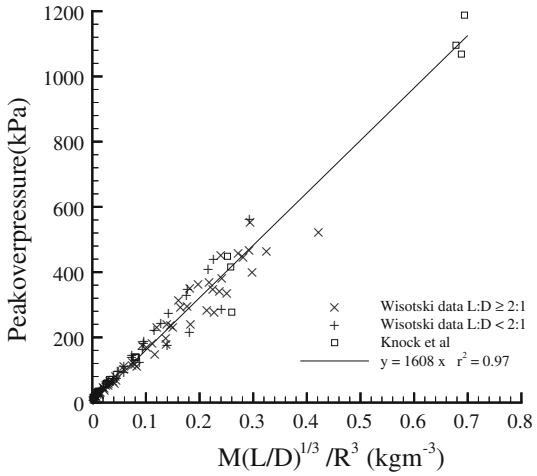


Fig. 3 Peak over pressure perpendicular to the curved surface of the cylinder plotted against $M(L/D)^{1/3}/R^3$ for Composition B. Data from Wisotski and Snyer 1965 [7] and the authors

At all distances from the charge, the impulse, I, can be predicted by either of [5]

$$I = K_I \frac{M^{0.5}}{R} \quad \text{or} \quad I = K_I' \frac{M^{2/3}}{R} \tag{5}$$

where K_I and K_I' are explosive-dependent constants and there is currently not enough data available to determine which equation is best.

2 Experimental work

The authors collected experimental data for bare charges of PE4 (RDX/binder 88/12) of 0.1–5 kg and *L/D* 4/1 to 6/1 and cast RDX/TNT 60/40 charges of 0.14 and 0.42 kg and *L/D* of 5/1 to 6.8/1. The PE4 charges were made by hand compaction of PE4 into cylindrical moulds lined with cardboard, from which the lightly cased charges could be removed without distortion. The charges were end initiated at 0° using commercial 8 star detonators that contain approximately 1 g of PETN. To avoid ground reflection the explosive charges of ≤400 g were placed at 2 m or more above the ground and the charges of ≥1 kg at 3 m above ground level (Fig. 2). The blast waves were measured side-on using Kistler pressure gauges level with the charge at 0, 45, 90, 135 and 180° to the charge.

3 Data analysis

Working with the literature data for Composition B [7] and Pentolite [8] and their own data for RDX/TNT 60/40 and PE4 the authors fitted equations to the data for peak over pressure and impulse in order to obtain equations predicting the values for a given charge mass, *L/D* ratio and distance. Ambient conditions affect blast parameters and the data are from two different heights above sea level and therefore ambient pressures; 83 kPa for Wisotski and Snyer and 102 KPa for the author's data. However, to avoid further adjustments to the raw data, no adjustment in the values was made for the height difference, e.g., by using Sachs scaling [14]. Only the peak over pressure and impulse perpendicular to the curved surface of the cylindrical charge are considered.

Table 1 Correlation coefficients for plots of peak over pressure perpendicular to the curved surface of the cylinder against mass, *L/D* and distance

Ratio	Composition B	Pentolite	PE4
M/R^3	0.82	0.82	0.95
$M/(R^3/(L/D)^{1/3})$	0.2	0.4	0.9
$M/(L/D)^{1/2}/R^3$	0.92	0.95	0.94
$M/(L/D)^{1/3}/R^3$	0.97	0.97	0.95
$M^{2/3}(L/D)^{1/9}/R$	0.84	0.89	0.87
$M/(L/D)^{1/4}/R^3$	0.97	0.96	0.95

3.1 Pressure

Using Composition B data from Wisotski and Snyer [7] for *L/D* ratios of 1/4 to 10/1 and mass 0.7—3.7 kg and the authors' own RDX/TNT 60/40 data for *L/D* ratios of 4/1 to 6.8/1 and mass 0.14–0.42 kg, the authors fitted lines to the data for peak over pressure. Curves of the form $P = M^n L/D^m/R^q$ were used, where n, m and q were constants varied to find the best root mean square fit to the data. It was found that the best fit (Fig. 3) was (Table 1) for

$$P = 1{,}608 \frac{M(L/D)^{1/3}}{R^3} = K_P \frac{M(L/D)^{1/3}}{R^3}. \tag{6}$$

If the results are split between the two sets of data, the Wisotski and Snyer data give $K_P = 1{,}604$ m^3 Pa kg^{-1} with $r^2 = 0.94$. The more limited authors' data gives

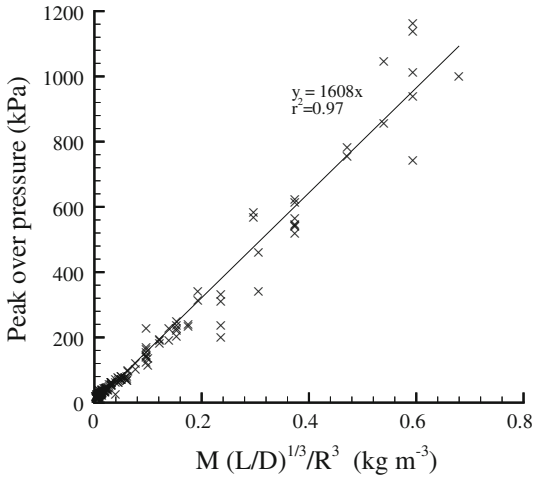


Fig. 4 Peak over pressure perpendicular to the *curved surface* of the cylinder plotted against $M(L/D)^{1/3}/R^3$ for Pentolite. Data from Plooster [8]

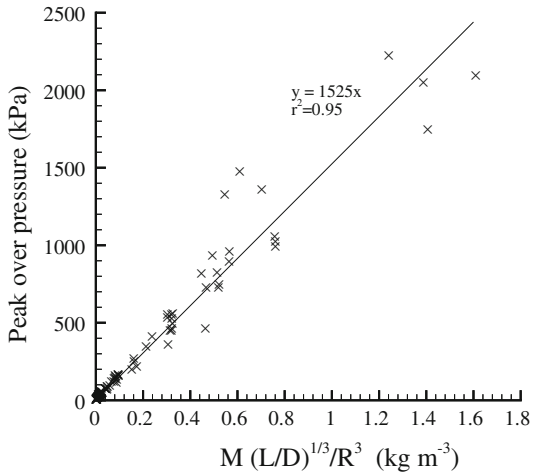


Fig. 5 Peak over pressure perpendicular to the *curved surface* of the cylinder plotted against $M(L/D)^{1/3}/R^3$ for PE4

$K_P = 1{,}612$ m^3 Pa kg^{-1} with a correlation coefficient, r^2, of $r^2 = 0.99$. The difference between the values of 0.5 % shows that for the data available there is no significant difference between the results, despite the differences in ambient conditions.

For Pentolite, the data in Plooster [8] is for *L/D* ratios of 1/4 to 6/1 and mass of 8 and 16 lb (3.63 and 7.26 kg). An equation of the form of (6) was found to give the best fit to the data (Table 1), with $K_P = 1{,}608$ m^3 Pa kg^{-1} (Fig. 4).

For PE4, the data are limited to *L/D* ratios of 4/1 to 6/1 and it was found that altering the power of *L/D* had little effect on the results. A wider range of data will be needed to determine the best data fit. For an equation of the form of (6), K_P is equal to 1,525 m^3 Pa kg^{-1}, Fig. 5.

Equation 6 states that for a given mass the peak over pressure in the direction perpendicular to the curved surface of the

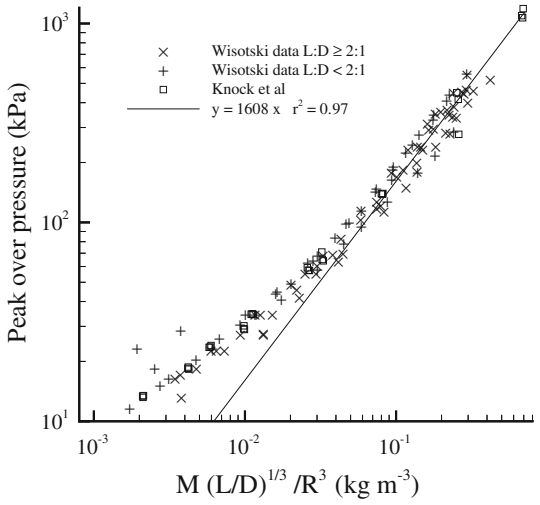


Fig. 6 Peak over pressure perpendicular to the *curved surface* of the cylinder plotted against $M(L/D)^{1/3}/R^3$ for Composition B using a log scale. Data from Wisotski and Snyer 1965 [7] and the authors

cylinder increases as the length increases relative to the diameter. This agrees with the observations by previous authors [7,10].

For *L/D* ratios of greater than 2/1 the authors found [4] that whilst at a scaled distance of $Z < 3.5$ m kg$^{-1/3}$ simple equations of the form of (3) could be fitted to the data, further out, as the cylindrical blast wave 'healed' to that of a sphere the more complex (4) was needed. When all length-to-diameter ratios are included, a similar effect is seen (Fig. 6), for Composition B and RDX/TNT 60/40. At a value of $M(L/D)^{1/3}/R^3 \approx 0.08$ kg m^{-3} , the simple empirical fit breaks down and a more complex equation is needed. Therefore, for a given mass of explosive, as *L/D* increases the distance at which the blast heals to that of a sphere increases. Fitting equations similar in form to (4) give the best fit of

$$P = C_1 \frac{1}{Z''3} + C_2 \frac{1}{Z''2} + C_3 \frac{1}{Z''} \tag{7}$$

where

$$Z'' = \frac{R}{M^{1/3}(L/D)^{1/9}} \tag{8}$$

and $C_1 = 1{,}858$ Pa kg m^{-3}, $C_2 = -402$ Pa kg$^{2/3}$ m^{-2}, $C_3 = 149$ Pa kg$^{1/3}$ m and the correlation coefficient is $r^2 = 0.97$. This is the same correlation coefficient as for (6) except now the variation of the fitted line from the data is across the entire spread of data, Fig. 7 and not just close in to the charge. The first term in (7) is of the same form as (6) for cylindrical charges, representing the blast close in and the third term is similar to the scaled distance used for spherical charges representing the blast once it has healed to that of a spherical charge. This leaves the central term which may represent the transition between the two phases. For Pentolite the constants are $C_1 = 2{,}500$ Pa kg m^{-3}, $C_2 = -999$ Pa kg$^{2/3}$ m^{-2},

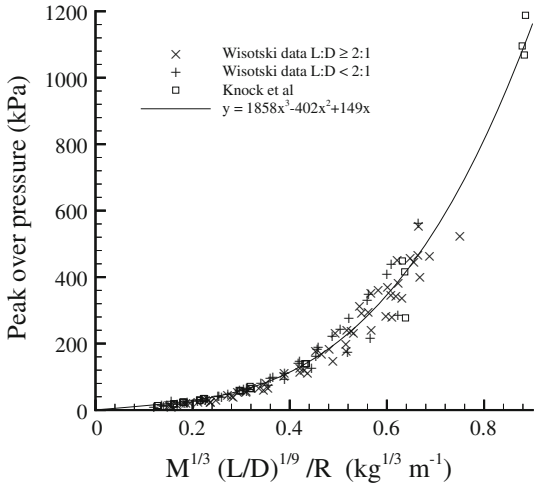


Fig. 7 Peak over pressure perpendicular to the *curved surface* of the cylinder plotted against $M^{1/3}(L/D)^{1/9}/R$ for Composition B. Data from Wisotski and Snyer 1965 [7] and the authors

Table 2 Correlation coefficients for plots of impulse perpendicular to the curved surface of the cylinder against mass, *L/D* and distance

Ratio	Composition B	Pentolite	PE4
$M^{0.5}/R$	0.91	0.82	0.87
$M^{0.5}(L/D)^{0.25}/R$	0.85	0.9	0.87
$M^{0.5}/(R(L/D)^{0.25})$	0.86	0.35	0.87
$M^{2/3}/R$	0.9	0.82	0.87
$M^{2/3}(L/D)^{1/9}/R$	0.84	0.89	0.87
$M^{2/3}/(R(L/D)^{1/9})$	0.87	0.67	0.87

$C_3 = 240$ Pa kg$^{1/3}$ m and the correlation coefficient is $r^2 = 0.97$ and for PE4 $C_1 = 831$ Pa kg m^{-3}, $C_2 = 791$ Pa kg$^{2/3}$ m^{-2}, $C_3 = -121$ Pa kg$^{1/3}$ m and the correlation coefficient is $r^2 = 0.96$.

3.2 Impulse

Using Composition B data from Wisotski and Snyer [7] and RDX/TNT 60/40 from the authors, it was found that the best data fit (Table 2) was still for equations of the form of (5), without taking into account the variation in *L/D*. For the authors' PE4 data with only a narrow variation in *L/D*, altering the *L/D* ratio made no significant difference to the results. In contrast, the pentolite data from Plooster [8] shows an improvement in the data fit when the *L/D* ratio is taken into account. The best fit occurred for $I = M^{0.5} - (L/D)^{0.25}/R$. The mixed results suggest that the impulse has either a limited or no dependency on the *L/D* ratio.

4 Conclusions and discussion

An analysis has been carried out of the literature data (for Composition B and pentolite) and data collected by the authors for RDX/TNT 60/40 and PE4 for peak over pressure and impulse perpendicular to the curved surface of a cylindrical charge for all length-to-diameter ratios.

Close in to the charge, that is for $M(L/D)^{1/3}/R^3 \approx 0.08$ kg m^{-3} or less, the peak over pressure was found to vary as $P = K_P M(L/D)^{1/3}/R^3$.

The relationship implies that the peak over pressure decays as $1/R^3$ with distance close into the charge. Similarity analysis of the production of blast waves from explosive charges demonstrates that the peak over pressure would decay as $1/R^n$ with distance [19], where $n = 1$ for slabs, 2 for cylinders and 3 for spheres assuming that the ambient pressure is negligible and that for cylindrical charges, the cylinder is infinitely long and is initiated instantaneously along the entire length of the charge. However, close into an end-initiated, finite-length cylinder, the side on shock wave forms a curve (Fig. 1) that appears more hemispherical in shape, explaining the $1/R^3$ dependency of the peak over pressure of the cylinders.

As the length of the cylinder increases, the peak over pressure perpendicular to the curved surface also increases. The volume, *V*, of a cylinder is given by $V = \pi D^2 L/4$ and the presented surface area, *A*, by $A = DL$. Combining these gives $A = (4M/\rho\pi)^{2/3}(L/D)^{1/3}$, where ρ is the density of the explosive. Hence for a given explosive density and mass the curved presented surface area is proportional to $(L/D)^{1/3}$, the same dependency as the peak over pressure. So as the length-to-diameter ratio alters, the peak over pressure varies directly as the presented area varies.

The peak over pressure is proportional to the scaled distance, *Z*, with values for the coefficient of proportionality, K_P, of 1,604 m^3 Pa kg^{-1} for Composition B, 1,608 m^3 Pa kg^{-1} for Pentolite and 1,525 m^3 Pa kg^{-1} for PE4. The values would be expected to vary between explosives due to their different chemistry which affects the detonation velocity, quantity of gas produced and hence the shape of the blast wave. The values in this paper suggest that pentolite and Composition B produce similar peak over pressures, whilst PE4 produces a lower peak over pressure for a given mass and distance from the explosive. The literature values for TNT equivalencies (the strength of an explosive relative to TNT) of the three explosives vary between authors, due to differences in experimental methods used, but are typically, for pressure, 1.11 for Composition B [15], 1.42 for pentolite [15] and 1.3–1.35 for PE4 [16]. These literature values imply that pentolite and PE4 are more powerful than Composition B. The differences in the predicted relative strengths may be due to the known issues in the accuracy of TNT equivalency values due to different measuring methods (sand blast test,

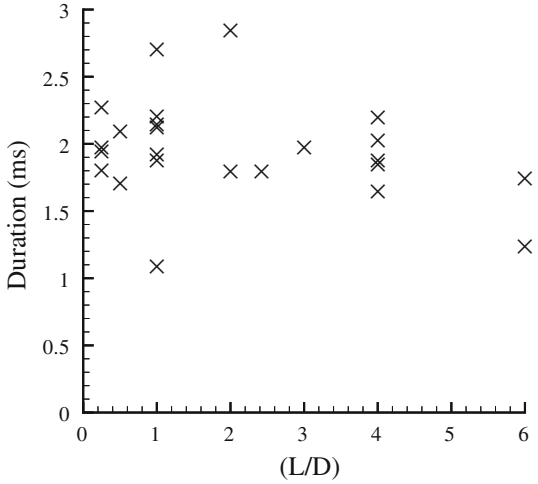


Fig. 8 Duration at 2.1 m (7 ft) perpendicular to the *curved surface* of the cylinder plotted against *L/D* for Pentolite. Data from Plooster 1982 [8]

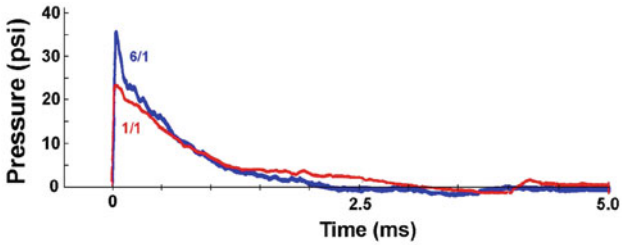


Fig. 9 Pressure–time *curve* for pentolite at 3.4 m (11 ft) from the charge for *L/D* = 6/1 and 1/1. Data adapted from Plooster 1982 [8]

bare spherical explosion, Trauzl test, etc., [17]), inconsistent measurement techniques and different explosives generating different-shaped blast waves [10,17,18]. Alternatively, the differences may signify a difference in the relative blast waves' shapes of the explosives due to the change in geometry from spherical to cylindrical charges.

Further out than $M(L/D)^{1/3}/R^3 \approx 0.08$ kg m^{-3} the blast wave 'heals' to that of a spherical blast wave and the peak over pressure at all distances can be obtained from an equation similar in form to [1] but taking into account the *L/D* ratio: $P = C_1/Z''^3 + C_2/Z''^2 + C_3/Z''$ where $Z'' = M^{1/3}(L/D)^{1/9}/R$. As stated by Stoner and Bleakney [1], this equation allows for the $1/R^3$ dependency close into the charge and the $1/R$ dependency far out from the charge.

The analysis has shown that the impulse from the curved surface of a cylindrical charge of explosive may not depend on the length-to-diameter ratio. If this is the case then given that the peak over pressure varies with the *L/D* ratio the shape of the blast wave must therefore alter with changing *L/D*. Data from Plooster [8] show that the duration of the wave does not vary with changing *L/D*, see Fig. 8. If the shape of the blast wave is studied (Fig. 9), then it can be seen that for large *L/D* the wave has a high peak over pressure but a quicker reduction in pressure than for lower values of *L/D*. This implies that for smaller *L/D* and hence larger diameter of the charge, there is more gas to expand outwards at a particular point along the cylinder, leading to a slower drop in pressure behind the peak over pressure. Hence, for a given mass of explosive the blast wave alters depending on the *L/D* ratio.

Explosives are compared using TNT equivalency, which compares, for example, the peak over pressure or impulse of one explosive to that of TNT. For cylinders close into the charge from the curved side of the cylinder this paper shows that the peak pressure can be predicted by $P = K_P M(L/D)^{1/3}/R$. As a result, a comparison of the values of K_P could give a method of developing a TNT equivalency for cylindrical charges.

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